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Acceleration of protons and deuterons up to 35 MeV and generation of 10^{13} neutrons in a megaampere deuterium gas-puff z-pinch

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Abstract

The acceleration of hydrogen ions up to 35 MeV is observed in the z-pinch experiments on the GIT-12 generator at a 3 MA current and 0.6 MV driving voltage. High ion energies are obtained with a novel configuration of a deuterium gas-puff z-pinch. In this configuration, a hollow cylindrical plasma shell is injected around an inner deuterium gas puff to form a homogeneous, uniformly conducting layer between electrodes at the initial phase of z-pinch implosion. The stable implosion at the velocity up to 650 km s^{-1} is important to deliver more current onto the z-pinch axis. Magnetohydrodynamic instabilities become apparent first at stagnation. After the disruptive development of $m = 0$ instabilities, $\sim 20 \text{ ns}$ pulses of high-energy photons, neutrons, electrons, and ions are observed. The average neutron yield is 2×10^{12} . The ion emission is characterized by various diagnostic techniques including those based on the usage of neutron-producing samples. When a large neutron-producing sample is placed onto the axis below a cathode mesh, the neutron yield is increased up to $(1.1 \pm 0.3) \times 10^{13}$. Considering a $\sim 130 \text{ kJ}$ energy input into z-pinch plasmas and magnetic field, this implies the neutron production efficiency of $\sim 10^8$ neutrons per one Joule of the z-pinch energy.

Keywords: z-pinch, gas puff, deuterium, cable guns, ion acceleration, neutrons

(Some figures may appear in colour only in the online journal)

1. Introduction

The z-pinch effect attracts a considerable interest due to its frequent occurrence in laboratory and space plasmas, and variety of applications. It was the simple principle and geometry in particular why z-pinch enjoyed great attention in the early 1950s in conjunction with controlled thermonuclear

fusion research [1]. The idea behind this research was to heat a fusion mixture by Joule heating and by adiabatic or shock compression and then to confine the plasma by the pinch effect until a sufficient amount of fusion energy was released. Indeed the first compressional z-pinch produced a high number of fusion neutrons, namely 10^8 per one pulse at a 300 kA current level [2]. At first, it was assumed that

deuterons were accelerated to fusion energies by multiple elastic collisions in high-temperature plasmas. However, shortly afterward, plasma physicists found out that the neutrons were not produced by Maxwellian plasmas. Instead, deuterons were supposed to be accelerated by axial electric fields during and after the growth of $m = 0$ instabilities [2–5]. The conclusion that neutrons were not of thermonuclear origin implied that z-pinchs could not be used for energy production. This led to the abandonment of a straight z-pinch as a fusion power source. Despite this, z-pinchs have remained an interesting research topic up to now.

At present, z-pinchs are often used to produce high-energy density ($>0.1 \text{ MJ cm}^{-3}$) plasmas. The utilization of z-pinchs is motivated by a high (10%–30%) conversion efficiency of stored electrical energy to plasma energy. Perhaps the best-known use of the high intrinsic efficiency of z-pinchs is the generation of soft x-ray radiation [6, 7]. This is why the primary interest of the z-pinch research was focused on a possibility to use z-pinch sources as x-ray drivers for inertial confinement fusion (ICF) in the 1990s and 2000s [8]. However, there has been always an attempt to exploit the z-pinch effect in a more direct thermonuclear fusion concept. Already in 1962, Linhart *et al* suggested that an imploding metallic shell could compress fusion plasmas and magnetic fields [9] (In this paper, the authors originally coined the term ‘inertial confinement’). Recently, the z-pinch implosion of a cylindrical shell has become a key part of so-called magnetized liner inertial fusion (MagLIF). The basic principle of MagLIF is to employ the benefits of magnetic fields and inertial confinement [10–12]. Such an approach relaxes the constraints on density and time scales that are necessary for laser-driven ICF. The first MagLIF experiments demonstrated fusion-relevant conditions including the magnetization of fusion products [13, 14]. Since then MagLIF belonged to three standard approaches to ICF in the USA.

As mentioned above, z-pinchs are susceptible to magnetohydrodynamic instabilities. These instabilities, particularly a symmetric $m = 0$ mode, led to the acceleration of deuterons and non-thermal neutron production in the first z-pinch and plasma focus experiments. Since stable implosion and stagnation are essential for high x-ray powers (e.g. in wire-array z-pinchs [15, 16]) and thermonuclear yields (e.g. in MagLIF [17, 18]), much effort of the z-pinch community has been devoted to suppressing all kinds of instabilities. In our research, we try to follow a different idea. We regard instabilities and particle acceleration as natural and fascinating z-pinch phenomena. Therefore, we do not want to suppress all instabilities in our experiments. On the contrary, we attempt to optimize the process that naturally occurs in z-pinchs. We suppose that z-pinchs could create powerful charged-particle beams if instabilities develop in an efficient way. This is precisely the main purpose of our deuterium gas-puff experiments that have been carried out on the GIT-12 generator at $\sim 3 \text{ MA}$ currents and $\sim \mu\text{s}$ rise times.

On GIT-12, we use a novel configuration of a deuterium z-pinch in which a hollow cylindrical plasma shell is injected around an inner deuterium gas puff [19, 20]. This configuration enables us to deliver more current onto the axis during

stable implosion whereas a symmetric instability develops later at stagnation. After the disruptive development of the $m = 0$ instability, a $\sim 20 \text{ ns}$ pulse of high energy photons, neutrons, electrons, and ions is observed [21]. Comparing our experimental results with state-of-the-art kinetic simulations [22], the typical neutron yields of 2×10^{12} on GIT-12 are more than one order of magnitude higher than predictions for 3 MA. Another uniqueness of the GIT-12 generator is that it is able to accelerate deuterons to an unprecedented energy exceeding 30 MeV [21]. This energy is about five times greater than the maximum deuteron energy ever observed in z-pinchs (see [23]). In addition to that, when we realize that 30 MeV is fifty times the ion energy provided by the GIT-12 driving voltage of 0.6 MV, these are exciting results which point to the potential of particle acceleration in z-pinchs and which deserve to be studied in more detail.

We published the first characterization of the ion beams on GIT-12 in [21]. Then we presented and discussed parameters which are essential for the identification of the ion acceleration mechanism in [24]. In this paper, we present further investigations of plasma dynamics and ion emission, particularly characterized via neutron-producing samples. The paper is organized as follows: the current generator, gas-puff hardware, experimental arrangement, and diagnostics used in our experiment are described in section 2. Section 3 provides the information about the plasma dynamics before, during, and after the ion and electron acceleration. The characteristic features of ion beams are presented in section 4. Section 5 describes the experimental results obtained with neutron-producing samples that were placed in the path of accelerated ions. The use of a large neutron-producing target led to a neutron yield increase which is the subject of section 6. Obtained results are discussed in section 7. Finally, conclusions are summarized in section 8.

2. Apparatus and diagnostics

2.1. Current generator

The experiments have been carried out on the GIT-12 generator at the Institute of High Current Electronics in Tomsk [25]. GIT-12 capacitors were charged to 50 kV and stored a total energy of 2.6 MJ. During the experimental campaigns described in this paper, the generator was used in a micro-second regime, i.e. without the plasma-opening-switch. In this regime, the current reached a peak of 4.7 MA with a current rise time of about $1.7 \mu\text{s}$ in a short-circuit load. In our experiments, gas puffs imploded onto the axis before the peak current at about 700 ns when the current reached 2.7 MA. As time zero we took a moment of a 100 kA load current.

2.2. Z-pinch load

High ion energies and efficient neutron production on GIT-12 were obtained with a hybrid deuterium gas-puff. In this configuration, an inner 8 cm diameter annular deuterium gas puff was surrounded by an outer hollow cylindrical plasma

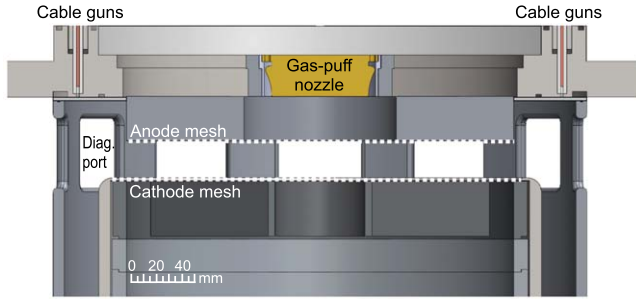


Figure 1. Experimental arrangement of the electrode system, gas-puff hardware and cable guns.

shell of a 35 cm diameter (see figure 1). Since 2013, we have performed 150 shots with various initial parameters. An optimal linear mass of the deuterium gas puff was about $100 \mu\text{g cm}^{-1}$. This linear mass was injected from a concentric convergent-divergent nozzle with a $1000 \mu\text{m}$ throat width and with a $300 \mu\text{s}$ time delay between the valve opening and the triggering of the generator. As shown in figure 1, the nozzles were placed on the anode side about 35 mm from the anode mesh. An anode-cathode gap varied between 20 and 28 mm in most of the shots. The diameter of the cathode was 32 cm, and the inner diameter of a return-current conductor approached 37 cm. Both the anode and cathode were formed by a stainless-steel mesh with a transparency of about 70%. The mesh consisted of 0.5 mm diameter wires. Due to multiple reflections of a fraction of the gas onto the mesh electrodes, the gas was spread out over a large area.

The main idea behind using the plasma shell was to form a homogeneous, uniformly conducting current sheath at a large initial radius prior to implosion (see [20] for more details). A hollow cylindrical plasma shell consisting of hydrogen and carbon ions was injected between electrodes by 48 polyethylene cable guns. The optimal conditions for high neutron yields were achieved when the cable guns were triggered $1.7\text{--}1.8 \mu\text{s}$ before the main z-pinch current. This timing provided a $\sim 5 \mu\text{g cm}^{-1}$ mass loading of the plasma shell inside the anode-cathode region.

2.3. Diagnostics

The hybrid deuterium gas-puff z-pinch on GIT-12 was studied by a comprehensive set of electrical, optical, x-ray, gamma, ion and neutron diagnostics, part of which has been already described in [20, 21, 26]. In the last experimental campaign in 2018, we have used:

- (i) High-voltage, current and dI/dt probes [25, 27].
- (ii) Three magnetic probes, one-turn loops with effective diameters of about 1.0 mm, placed at 0.6 mm above the cathode mesh at 6, 9 and 13 cm from the z-pinch axis.
- (iii) An optical streak camera operating in the radial mode. The emission from the plane at $10.0 \pm 0.2 \text{ mm}$ above the cathode was imaged on the slit of the streak camera.
- (iv) A four-frame soft x-ray pinhole camera of 2 ns exposure and 0.53 magnification filtered by a $0.7 \mu\text{m}$ thick Mylar film.

- (v) A time-integrated x-ray pinhole camera with a $10 \mu\text{m}$ Be filter and 1.004 magnification.
- (vi) An x-ray vacuum diode with a copper cathode filtered by a $1.4 \mu\text{m}$ Mylar foil (the peak sensitivity in a 150–280 eV window).
- (vii) Two photoconducting detectors with $20 \mu\text{m}$ polypropylene and $30 \mu\text{m}$ teflon filters detecting the bremsstrahlung radiation above 1.7 keV.
- (viii) A silver activation counter located radially at 4.55 m for the measurement of neutron yields.
- (ix) Aluminum, copper, indium, zirconium and other samples placed at various directions and distances for the neutron fluence measurement above specific threshold energies. A NaI:Tl scintillator detector and a high-purity Ge detector for the post-shot gamma spectroscopy of samples activated by neutrons and fast ions.
- (x) A RDS-31S gamma radiation detector measuring the time evolution of a dose-rate caused by 0.06–6 MeV photons at a 1 m distance from the center of the experimental chamber.
- (xi) Four radial and one axial (downstream) neutron time-of-flight (TOF) detectors [28] placed between 2.00 and 25.8 m from the z-pinch. For preventing bremsstrahlung radiation from saturating the photomultipliers, the detectors were shielded by 15–20 centimeters of lead and neutral density filters were placed between a BC-408 scintillator and a Hamamatsu photomultiplier tube.

Since 2014 the emphasis has been put on the diagnostics of ions and electrons emitted in the axial direction towards the cathode and anode, respectively. For this purpose, we placed various ion detectors (beam-profile detector, multi-pinhole cameras, neutron-producing samples, etc) below the cathode mesh and we measured angular, spatial, spectral, and temporal properties of emitted hydrogen ions in the 1–40 MeV range (see e.g. [21, 24]). The parameters of electron beams were characterized by detectors placed inside gas-puff nozzles (pinhole camera, beam-profile detector, magnetic spectrometer, thermoluminescence dosimeters). More details about the diagnostics will be provided in the following sections where appropriate.

3. Gross dynamics of a hybrid deuterium gas-puff z-pinch

Dynamics of z-pinch implosion, stagnation, and disruption was studied particularly with the four-frame x-ray camera and the optical streak camera operating in the radial mode. Figure 2 presents the exemplary images obtained in three shots. Time zero is defined as the sharp rise of the γ -ray pulse.

In figure 2, the first two images at -15 and -5 ns show the final phase of deuterium gas-puff implosion which was quite stable and reached the radial velocity of $v_{\text{imp}} = 6.5 \times 10^5 \text{ m s}^{-1}$. This velocity corresponds to an ion temperature $kT_i = \frac{1}{3} m_d v_{\text{imp}}^2 = 3 \text{ keV}$ at stagnation. The stagnation, if ever occurred, lasted only several nanoseconds. Shortly after the end of implosion, the z-pinch column was disrupted by $m = 0$

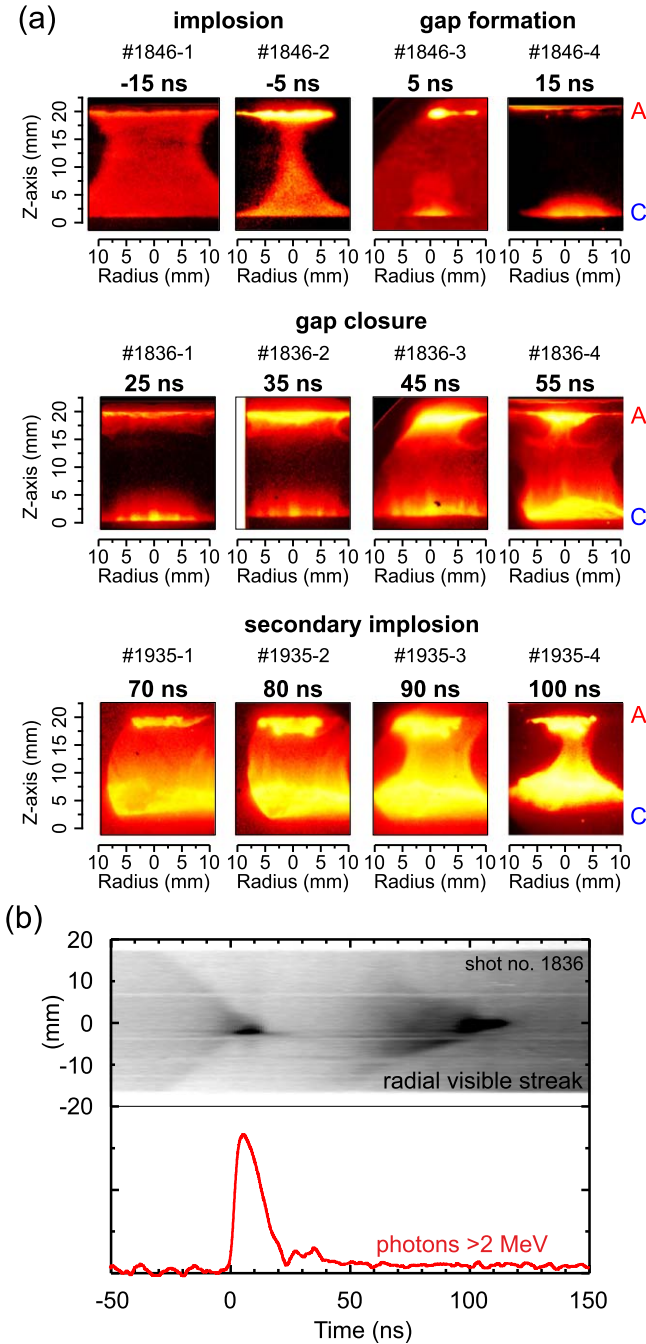


Figure 2. (a) Soft x-ray pinhole images from three z-pinch shots with a hybrid deuterium gas-puff (No. 1836, 1846 and 1935). The anode is at the top and cathode is at the bottom. Time $t = 0$ is defined by a sharp onset of a γ -ray pulse. (b) Radial streak-camera image and high-energy bremsstrahlung radiation in shot no. 1836.

instabilities (see images in [20]). The third image at +5 ns demonstrates that the gap was formed first near the anode. Later, at +15 ns, the emitting plasmas could be seen only in the vicinity of electrode meshes. The disruption was accompanied with the principal neutron pulse and high energy (> 2 MeV) bremsstrahlung radiation as shown in figure 2(b). The increase of x-ray background radiation is also visible in the third image of figure 2(a) at +5 ns. High-energy bremsstrahlung radiation and neutron emission lasted 20 ns. During

this period, on average, 2×10^{12} neutrons were produced. The peak neutron yield observed in 150 shots with the hybrid deuterium gas-puff, i.e. without any neutron-producing catcher, was $(5.9 \pm 1.2) \times 10^{12}$. In order to be confident with the origin and absolute yields of neutrons, the neutron emission was diagnosed by several independent techniques described in [29].

Figure 2 further shows the evolution after the gap formation. Between +20 and +50 ns, the plasma expansion from both electrodes and the anode–cathode gap closure were observed. 50 ns after the disruption, the secondary implosion with the velocity of about 3×10^5 m s $^{-1}$ occurred. The delay between the first and the second stagnation was about 100 ns and, interestingly, it was shorter in the shots with higher neutron yields. After the secondary implosion, neither hard x-rays nor neutrons were generated. The intensive pulses were observed mainly in the extreme-ultraviolet and soft x-ray region. Powerful soft x-ray radiation was produced also at the peak current. However, we have not studied the plasma dynamics during the peak current in more details since we were interested mainly in fast electrons and ions that were accelerated during the disruption. As far as the ion emission is concerned, we will describe its characteristic features in the following section.

4. Characteristics of accelerated ions

A hybrid deuterium gas-puff z-pinch has proven to be not only an efficient source of neutrons but also an interesting accelerator of hydrogen ions. In the previous work [21], we used various ion diagnostic instruments to characterize the ion emission. Since gas-puff z-pinch produces specific ion beams in each shot, it is crucial to use comprehensive ion diagnostics which provides at the same time the information about angular, spatial, and spectral properties of an ion source. For this purpose, we constructed an axial detector which is displayed in figure 3(a). The exemplary results obtained with this detector are displayed in figures 3 and 4. Two observations seem to be characteristic for the ion emission from our deuterium gas-puff z-pinch. The first characteristics is the detection of radial lines with an ion beam profile detector (see figure 3). The second one is a ring structure which is observed by an axial ion pinhole camera (see figure 4). We will focus on these two characteristics in the following paragraphs.

Figure 3 shows images that were detected with a CR-39 solid-state nuclear track detector and radiochromic films below the cathode mesh. These detectors were placed at 10 cm below the cathode mesh to measure the profile of an ion beam. Usually, the beam profile consisted of many radial lines more or less symmetrically distributed around the center of a beam. We explain the origin of these radial lines by collimated ion beams which were deflected by azimuthal magnetic fields or, alternatively, by radial electric fields [24]. The analysis of the tracks at the CR-39 detector confirmed that the radial lines were created by accelerated protons or deuterons. In the shot presented in figure 3(d), hydrogen ions

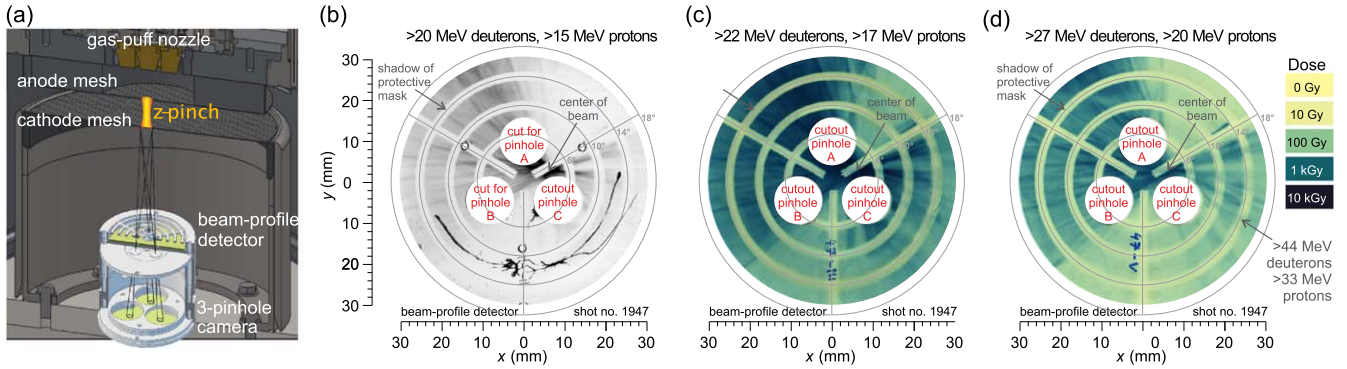


Figure 3. Measurement of ion emission anisotropy. (a) Scheme of the axial ion detector. (b)–(d) Images of the ion-beam profile detected by a CR-39 detector (b) and the second and third HD-V2 gafchromic films (c)–(d) at 10 cm below the cathode mesh. Shot no. 1947. The stack consisted of a 1 mm duralumin cover, a 1 mm stainless-steel protective mask supported with 1 mm rubber, 0.1 mm aluminum absorber, the first HD-V2 film, 0.5 mm CR-39 detector, the second HD-V2 film, 0.5 mm aluminum absorber, the third HD-V2 film, 0.5 mm CR-39 detector. The energies of the detected ions were calculated with the SRIM code [46].

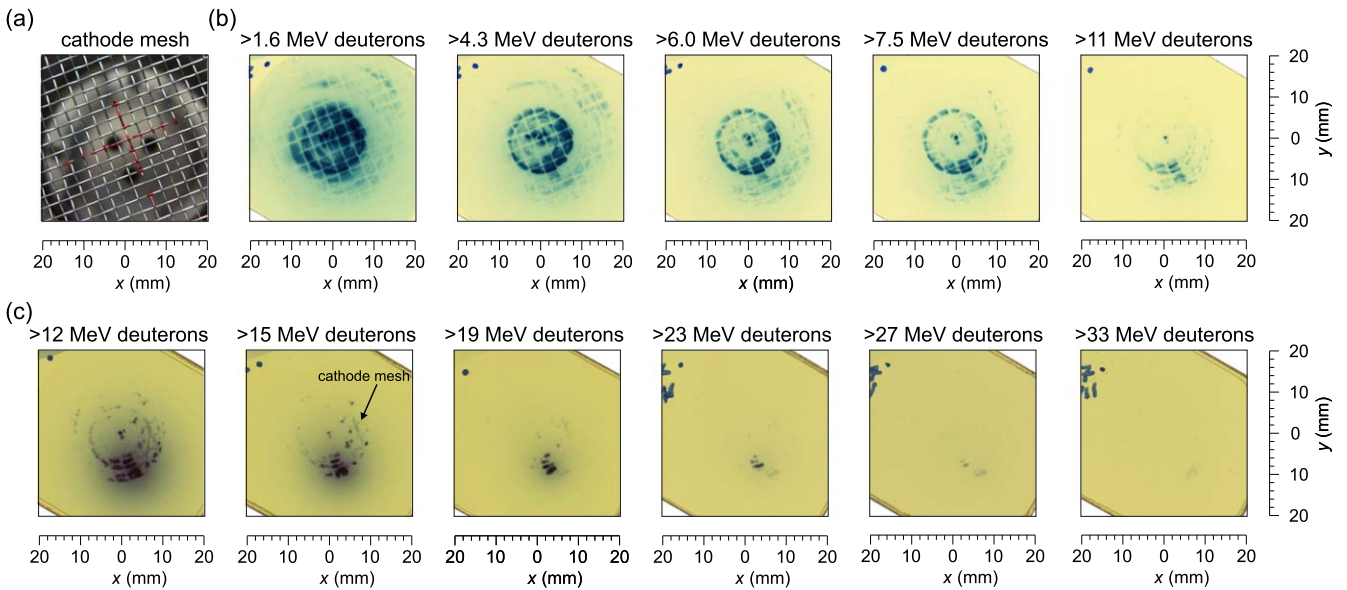


Figure 4. (a) On-axis view of the cathode mesh with the axial ion detector below it. (b) Images from the ion pinhole camera (pinhole C) recorded by the stack of HD-V2 films. (c) Images from the ion pinhole camera (pinhole C) recorded by the stack of EBT-3 films. Spatial scales correspond to the plane of the cathode mesh. Pinhole camera had the magnification of 0.55 and pinholes of a 0.26 mm diameter. Shot no. 1947.

reached very high energies above >20 and >27 MeV in the case of protons and deuterons, respectively. In addition to that, we used a 1 mm thick stainless steel mask together with 1 mm thick rubber to additionally protect the detectors from discharge debris. Since the radial lines are clearly visible also behind this additional mask, the energies of hydrogen ions were undoubtedly above 33 MeV.

High ion energies were also recorded with the three-pinhole camera that was located below the beam-profile detector (see figure 3(a)). Figure 4 shows the result from pinhole C which was in shot no. 1947 close to the center of the ion beam. As can be seen in the ion pinhole images, hydrogen ions were recorded by all detectors in the stack. It corresponded to end-point energies of protons and deuterons above 25 or 33 MeV, respectively. The most energetic ions were emitted from individual localized spots. At lower

energies, the ions originated from a larger area, and the shadow of the cathode mesh was clearly visible. Interestingly, in several cases, we saw ions to be emitted from the cathode mesh (see e.g. pinhole images of >12 and >15 MeV deuterons in figure 4(c)). Since the ion emission from the cathode mesh was recorded simultaneously by more pinhole cameras, we explain it by the Rutherford scattering of high-energy ion beams. On the one hand, one would expect that the cathode mesh always reduces the energies and numbers of detected ions. On the other hand, however, due to the scattering of ions to large solid angles, the mesh could also cause the detection of ions that would otherwise not be recorded. For instance, in our case, the radial lines at the beam-profile detector (see figure 3) indicate that ions emitted from individual spots do not need to strike each pinhole and to be detected by pinhole cameras. However, if an intense high-energy ion beam is

scattered at the stainless-steel mesh, then the ions can be detected and the cathode mesh may seem to be a source of these ions.

The most characteristic features of ion pinhole images are ring-like structures observed mainly in the 1–10 MeV range. In [24], we attempted to distinguish whether the annulus represents a real spatial distribution of an ion source or it is created by magnetic fields after ion acceleration. This distinguishing is crucial for the estimation of a real ion-source diameter and thus for the discussion of ion acceleration mechanisms. Here, we would like to draw attention to another fact which should be taken into account. Regardless the original spatial distribution of ions and the effect of magnetic fields, the pinhole images in figure 4 show the spatial distribution of the ion flow in the plane of the cathode mesh. The non-distorted shadow of the mesh confirms the hypothesis of the ballistic transport of ions in a low-pressure gas below the cathode mesh. On the basis of the gained pinhole images, one could conclude that the number of ions actually emitted right from the z-pinch center is not significant. However, it is necessary to realize that an individual ion pinhole camera detects the number of ions emitted from a place at the cathode mesh (r, φ) into a specific solid angle ($\Omega, \Omega + d\Omega$). It means that an ion pinhole camera measures $dN(r, \varphi, \Omega)/d\Omega$. In order to estimate the number of ions emitted from any place on the cathode mesh $N(r, \varphi)$, we have to integrate over the solid angle at which ions are emitted. Since on-axis and off-axis ion beams might have completely different divergences, the result from one ion pinhole camera could be misleading if we are interested in the total number of ions. In order to estimate the spatial distribution of ions emitted in the plane of the cathode mesh $N(r, \varphi)$, we have to use a different technique which will be described in the following section.

5. Neutron emission from LiF samples placed at various distances from the z-pinch axis

One possibility how to characterize the solid angle at which ions are emitted is to use more pinhole cameras under different lines of sights. For this purpose, we designed a nine-pinhole camera. Even though our nine-pinhole camera demonstrated that on-axis ion emission had larger emission divergence than off-axis ion beams, we were not able to obtain more quantitative values due to a limited number of pinholes. Therefore we decided to estimate the relative number of ions via neutron-producing samples placed right below the cathode mesh. To distinguish primary DD neutrons from secondary neutrons produced in a sample, we used lithium fluoride with a natural abundance of ^7Li isotopes. Figure 5 presents the signals from neutron ToF detectors placed at 2.00 and 25.79 m from the z-pinch. A characteristic >10 MeV neutron peak from the $^7\text{Li}(d, n)$ reaction was nicely visible at the distant detector.

Unlike our previous work [21], LiF samples of a 1 mm thickness were placed at various distances from the z-pinch axis, namely at 0, 3, 6, 10, 12, 16 and 26 mm. The size of LiF samples was chosen to produce a distinguishable number of neutrons from the $^7\text{Li}(d, n)$ reaction. The position of LiF samples with respect to the center of an ion beam was

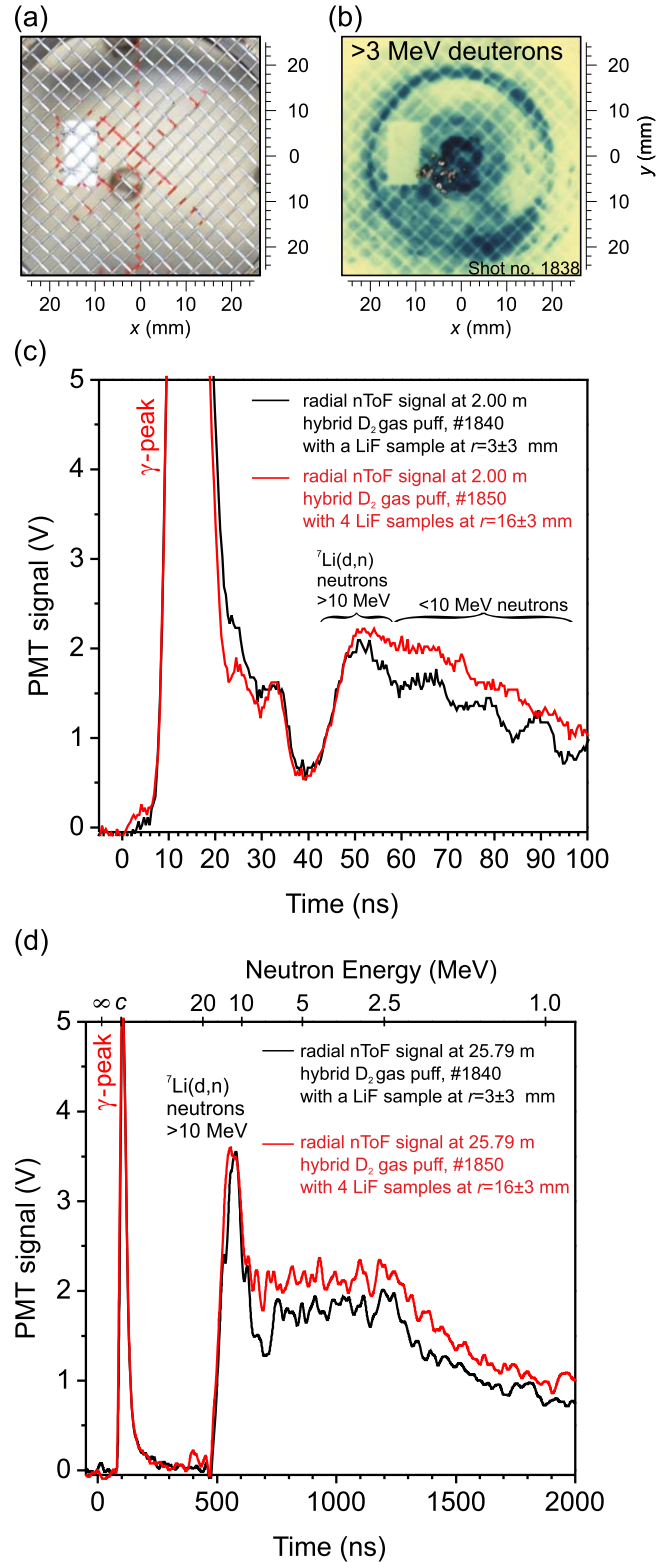


Figure 5. (a) Photo and (b) ion-pin-hole image of the cathode mesh and the off-axis LiF sample in shot no. 1838. (c), (d) Radial neutron ToF signals at 2.00 m (c) and 25.79 m (d) from a deuterium gas-puff z-pinch with LiF samples at 3 mm (black lines) and 16 mm (red lines), respectively. Shots no. 1840 and 1850, respectively.

estimated on the basis of ion pinhole images (see figure 5(b)). Figures 5(c) and (d) show the neutron ToF signals when LiF samples were placed at 3 and 16 mm from the z-pinch axis.

Despite the different distance from the axis, we did not see any difference in the onset of neutron ToF signals at 2.00 m (the nearest detector for the evaluation of neutron production time) and at 25.79 m (the most distant detector for the measurement of neutron energy spectra). Figure 5, therefore, indicates that ${}^7\text{Li}(d, n)$ neutrons at 3 and 16 mm started to be produced simultaneously (within a nanosecond precision) and that the energy spectra of reacting deuterons were not significantly different. Due to the similar spectra of neutron-producing deuterons, we can assume that the neutron yield from LiF samples is directly proportional to the number of accelerated ions at the position a LiF target.

The yields of >10 MeV and >12 MeV neutrons were measured by a BDS-10000 super-heated fluid detector [30] and by the ${}^{63}\text{Cu}(n, 2n){}^{62}\text{Cu}$ reaction in the copper activation sample [29], respectively. Most of >10 MeV neutrons originated from the ${}^7\text{Li}(d, n)$ reaction with the Q -value of 15.03 MeV. However, the residual nuclei of this reaction often remain in excited states and three-body processes, e.g. ${}^7\text{Li}(d, n+\alpha){}^4\text{He}$, contribute to neutron production. In our calculations, we therefore assumed that >10 MeV neutrons formed approximately a quarter of the neutron yield produced by the LiF catcher (see e.g. [31]). This way we obtained the LiF neutron yields and results presented in figure 6.

Figure 6(a) shows the dependence of neutron yields from a 1 cm^2 LiF sample on its radial position. This dependence was obtained in the series of shots with relatively high reproducibility. The reproducibility of shots was monitored through relevant parameters such as a peak voltage, the dip in dI/dt , high-energy bremsstrahlung radiation, ion energies, a total neutron yield, etc. As shown in figure 6(a), the neutron yield seemed to be an exponentially decreasing function of a radial distance. Evidently, on-axis samples placed below the cathode mesh produced significantly more neutrons than off-axis catchers. At larger radii, however, it is possible to produce more neutrons by a larger LiF sample. If we take the factor of $2\pi R$ into account, we obtain the dependence of a differential neutron yield as displayed in figure 6(b).

Figure 6(b) shows that the neutron yield from a LiF sample of a 1 mm width and $2\pi R$ length is about 2×10^{11} neutrons mm^{-1} up to a 12 mm radius. Then the differential neutron yield is decreasing. Integrating the dependence in figure 6(b) up to 12 mm, the total neutron yield produced by a LiF sample of a 24 mm diameter would be 2.2×10^{12} . This value is in rough agreement with the neutron yield of $(2.0 \pm 0.5) \times 10^{12}$ which was produced by using a $2.8 \times 2.8\text{ cm}^2$ square LiF sample [21]. Further, it should be mentioned that we placed LiF samples also on the anode mesh and gas-puff nozzles. Despite several attempts, we did not see any sign of a >10 MeV neutron peak. It means that the neutron yield from the $\text{Li}(d, n)$ reactions was at least two orders of magnitude lower than in the case of LiF placed below the cathode mesh.

In order to provide additional information about accelerated ions, we performed two shots with $3 \times 3\text{ mm}^2$ square

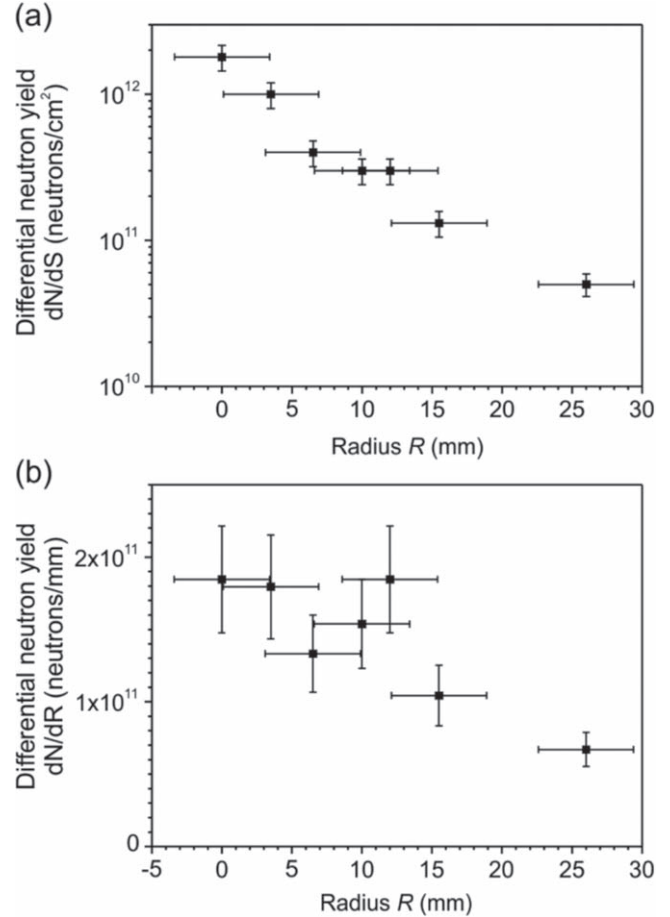


Figure 6. (a) Neutron yield from a LiF sample placed at various radial positions (normalized to 1 cm^2). (b) Neutron yield from a LiF sample of a 1 mm width and a $2\pi R$ length placed at radius R .

LiF samples covered by a $20\text{ }\mu\text{m}$ thick aluminum foil and placed on the z-pinch axis right below the cathode mesh. With the aluminum foil, the yield from the $\text{Li}(d, n)$ reaction decreased roughly by a factor of 2. Since the $20\text{ }\mu\text{m}$ thick aluminum foil fully absorbs deuterons up to a 1.5 MeV energy, it is evident that a significant neutron yield from LiF was produced by >2 MeV deuterons. Figure 7(a) presents the thick-target yields for reactions of deuterons with deuterated polyethylene, lithium fluoride (unfiltered and filtered with aluminum) and plasma target (2 cm length and $5 \times 10^{19}\text{ cm}^{-3}$ density). The two-fold decrease of the LiF thick-target yield with the $20\text{ }\mu\text{m}$ Al foil corresponds to the effective deuteron energy of 3.5 MeV.

6. Neutron yield increase

Figure 7(a) brings us to the idea of neutron yield increase by using deuterated polyethylene or lithium fluoride catchers. As shown in figure 7, deuterium plasma is a suitable target for deuterons with energies on the order of 100 keV. At this energy, the DD fusion cross-section is significant, and the ion

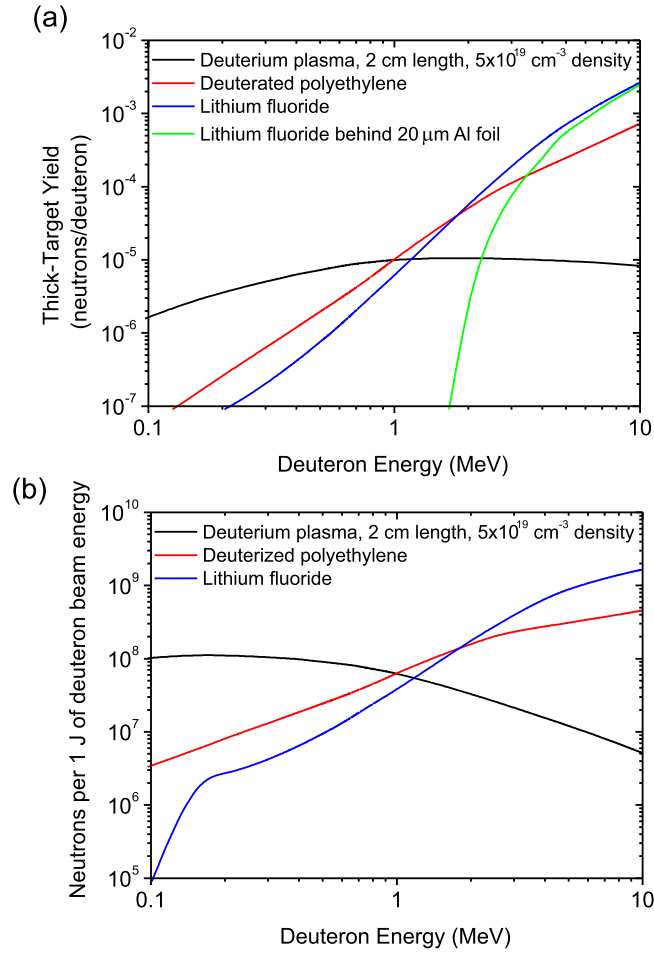


Figure 7. (a) Thick-target neutron yields for reactions of deuterons with deuterium plasmas of a 2 cm length and $5 \times 10^{19} \text{ cm}^{-3}$ density (black line), deuterated polyethylene (red line), natural lithium fluoride (blue line), and filtered natural lithium fluoride (green line). Thick-target yields were calculated with the ENDF and SRIM databases [46, 47] using the procedure described in [48]. The change of ion stopping powers in high-temperature (partially ionized) samples is not taken into account. (b) Number of neutrons produced by the deuteron beam with a total kinetic energy of 1 J.

stopping power in z-pinch plasmas is usually small. The efficient neutron production can be achieved particularly when ions are magnetized [19, 20, 32]. Above 100 keV, however, the fusion cross-section does not increase significantly and even decreases for energies above 2 MeV. If the z-pinch is able to accelerate multi-MeV deuterons, such as in our case, it is possible to increase the neutron yield by using neutron-producing samples. On the basis of our experimental results, we designed a large neutron-producing catcher which was described in [24]. As shown in figure 8(a), a 35 mm diameter and 0.4 mm thick deuterated-polyethylene (CD_2)_n disk was placed below the cathode mesh. On the axis, we used a small, 1 mm thick LiF sample which was more suitable for >2 MeV deuterons (see figure 7). The ion pinhole images, e.g. in figure 8(b), showed that ions reached relatively high energies and interacted with the neutron-producing samples.

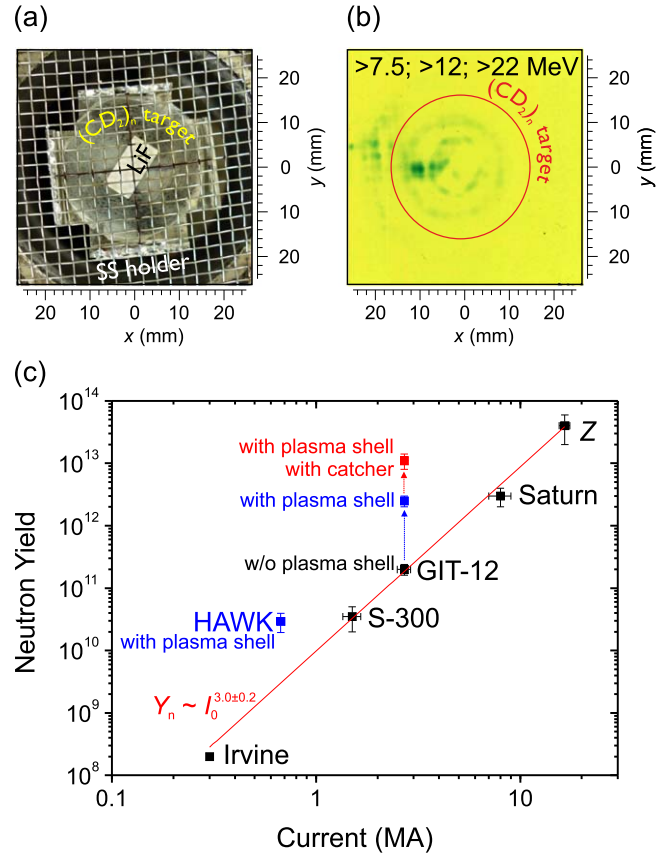


Figure 8. (a) Photo and (b) ion-pinhole image of a neutron-producing sample. The values of >7.5 , >12 and >22 MeV represent the deuteron energies needed to penetrate (i) 0.02 mm Al foil and three HD-V2 detectors; (ii) 0.1 mm stainless-steel foil, 0.02 mm Al foil and three HD-V2 detectors; (iii) 1 mm LiF catcher, 0.4 mm (CD_2)_n sample, 0.1 mm SS foil, a 0.02 mm Al foil and three HD-V2 detectors, respectively. (c) Neutron yields from deuterium gas-puff Z-pinch experiments [32, 49–51]. The increases of neutron yields on GIT-12 by using plasma shell [20] and neutron-producing catcher (this work) are displayed with blue and red arrows, respectively.

An intense central ion beam with the end-point energy above 22 MeV really hit the LiF sample. Also, the circle of >12 MeV deuterons interacted with the deuterated polyethylene disk. This way, we achieved a high neutron yield of $(1.1 \pm 0.3) \times 10^{13}$.

The significant neutron-yield increase with the large neutron-producing sample can be seen in figure 8(c) which displays the neutron yields on GIT-12 in 3 different configurations. The black point represents the shot with a standard D₂ gas-puff z-pinch [26] which fits quite well to other deuterium gas-puff z-pinch experiments without plasma shell (black points). The blue point on GIT-12 shows the shot when a plasma shell was injected around a gas puff [20]. Evidently, the yield was higher by an order of magnitude. Finally, the red point represents the last improvement, i.e. 10^{13} neutrons when the large neutron-producing sample was placed below the cathode mesh.

7. Discussion

7.1. Uniqueness of the hybrid deuterium gas-puff on GIT-12

The previous sections demonstrated that the hybrid deuterium gas-puff z-pinch on the GIT-12 generator achieved several unique parameters and results. For instance, we can mention the z-pinch implosion that was without significant magneto-Rayleigh–Taylor distortions despite a large radius of the outer plasma shell. In the case of gas-puff z-pinch, the implosion from a large initial radius is usually unstable. A relatively stable implosion is observed in dense plasma foci with a static deuterium gas filling and snowplow stabilization. When compared with dense plasma foci, the gas-puff z-pinch on GIT-12 provided a higher implosion velocity. Typical radial velocities in DPFs are about 250 km s^{-1} [33]. The implosion velocity in our experiment reached 650 km s^{-1} which implies a higher, fusion-relevant temperature of deuterons. Even though some ions were supposed to be accelerated to fusion energies by multiple elastic collisions during stagnation [34, 35], a significant thermonuclear production was avoided by the short stagnation and the fast disruption of a current-carrying column.

During and after the disruption, high-energy electrons, ions, photons, and neutrons were observed. By injecting the outer plasma shell, the neutron yields were increased by one order of magnitude to an average value of 2×10^{12} . This is how the ‘saturation’ of neutron yields between 10^{11} and 10^{12} observed for 30 years on megaampere devices (e.g. [36]) has been overcome. Another uniqueness of our mega-ampere z-pinch is that it was able to accelerate hydrogen ions to unprecedented energies exceeding 30 MeV. This is fifty times the ion energy provided by the generator driving voltage of 0.6 MV and also the highest energy observed in z-pinch and dense plasma foci. It should be stressed that, to achieve these energies, our experiment required neither a >10 megaampere current nor a modern multi-megavolt pulsed power generator. It is therefore natural to ask what makes the experiment on GIT-12 unique.

The uniqueness of the GIT-12 experiment can be discussed on the basis of the ion acceleration mechanism presented in our previous paper [24]. The mechanism described in [24] is based on Trubnikov’s idea of the disruption of a conduction current [37, 38]. In our case, a significant current drop is attributed to a transition from a low-impedance plasma to a space-charge limited flow (or self-magnetic insulation) in a gap formed after the ejection of plasmas from $m = 0$ constrictions. This behavior is analogous to the main principle of plasma-filled diodes [39–42] and plasma-opening switches [43, 44] where the gap formation in plasma is used for power and voltage multiplication. From the point of view of an equivalent circuit, ions are supposed to be accelerated by the voltage generated during a fast drop of the electric current, i.e. $V \approx -L dI/dt$. Here, L is the inductance which corresponds to the disrupted current-carrying column and to the magnetic energy stored around the z-pinch $\frac{1}{2}LI^2$. In order to reach the high induced voltage V , it is desirable to maximize the inductance and to disrupt the z-pinch current as fast as

possible. Based on this knowledge, we can see the advantage of GIT-12 in a high impedance (0.12Ω), or more precisely in a high output inductance (110 nH) of the generator. The high inductance allows us to use a relatively long z-pinch load and to have a return current cage of a 37 cm inner diameter without decreasing significantly the current at stagnation. Consequently, it means a higher inductance for a given final z-pinch radius. In addition to that, the final z-pinch radius was positively influenced by the usage of the cable guns. As shown in [20], the injection of the outer plasma shell around the inner deuterium gas puff led to more current delivered onto the axis. The smaller z-pinch diameter and fast implosion might be the cause of a fast current disruption along a significant part of the z-pinch length.

The hypothesis mentioned above is supported by recent experiments on the HAWK generator at the US Naval Research Laboratory in Washington. HAWK is also a high-inductance (607 nH) generator delivering a 0.64 MV, 0.67 MA pulse to a load within a $1.2 \mu\text{s}$ rise time. The HAWK generator is of a similar pulsed power architecture as GIT-12. HAWK could, therefore, provide not only new experimental data from complementary diagnostics but particularly unique information about the scaling of charged particle acceleration and neutron production. Recently, the preliminary experiments focused on DD neutron production have been carried out. The highest neutron yields were achieved when a deuterium plasma shell was injected from Marshall guns around a deuterium gas puff [45]. The average yield measured by a rhodium-activation counter was estimated as $(3 \pm 1) \times 10^{10}$. On the one hand, this yield is almost two orders of magnitude lower than on GIT-12. On the other hand, however, figure 8 shows that the yield on HAWK is well above the scaling law derived from ‘standard’ gas-puffs without using plasma shells. The gained yield, therefore, supports the hypothesis that the high output impedance of a generator and a plasma shell at a larger radius could be favorable for ion acceleration and neutron production.

7.2. Neutron production efficiency

In the previous sections, we mentioned that the hybrid deuterium gas-puff produced a large number of neutrons, mostly from the DD fusion reaction. Higher neutron yields and efficient neutron production in deuterium z-pinch plasmas were explained previously by the magnetization of MeV deuterons [20]. More recently, the better knowledge about multi-MeV ions has led us to the idea that additional neutron production can be achieved with a neutron-producing sample. When the large neutron-producing catcher from lithium fluoride and deuterated polyethylene was placed below the cathode mesh, the neutron yield was increased up to $(1.1 \pm 0.3) \times 10^{13}$. At this point it seems interesting to calculate the neutron production efficiency in terms of number of neutrons per one Joule of energy input into z-pinch plasmas and magnetic fields. In good shots, the energy input into the imploding z-pinch up to the end of neutron emission was $W_{\text{zpinch}} = \int V_{\text{zpinch}} I dt \approx 130 \text{ kJ}$. Consequently, for $(1.1 \pm 0.3) \times 10^{13}$ neutrons, we obtain the neutron production efficiency of $\sim 10^8$ neutrons per one Joule of ‘z-pinch’ energy. This is fairly high efficiency when compared

with a maximal theoretical prediction. The maximal theoretical prediction can be calculated if we assume that all the plasma and magnetic energy stored in a z-pinch is converted to deuteron beams. These deuteron beams can generate neutrons in various targets. Figure 7(b) shows the neutron yields produced by the interaction of a 1 J deuteron beam with z-pinch plasmas or with thick targets from deuterated polyethylene and lithium fluoride. In the range between 0.1 and 10 MeV, the deuteron beam of a 1 J total energy produces neutron yields between 10^8 and 10^9 . In our experiment, we reached the value of 10^8 per one Joule of z-pinch energy. It follows that, during the z-pinch disruption, the plasma and magnetic energy was efficiently (on the order of 10%) converted into ion beams and that we were close to the theoretical limit.

8. Conclusions

Z-pinches are known to be highly susceptible to magneto-hydrodynamic instabilities. Since stable implosions and stagnations are essential for high x-ray powers and thermonuclear yields, much effort of the z-pinch community has been devoted to suppressing all kinds of instabilities. In our research, we regard some instabilities and particle acceleration as natural and fascinating z-pinch phenomena. Therefore, we attempt to optimize the process which naturally occurs in z-pinches. For this purpose, we have carried out the z-pinch experiments with the hybrid deuterium gas puff on the GIT-12 generator at a 3 MA current and $\sim \mu\text{s}$ rise time. In this configuration, the implosion at the velocity up to 650 km s^{-1} seemed to be stable. Large-scale instabilities became more pronounced during stagnation at a 5 mm radius. When the disruption of necks occurred, high energy bremsstrahlung radiation and also the main neutron pulse of a $\approx 20 \text{ ns}$ FWHM were observed. The average neutron yield was 2×10^{12} . The ion emission was characterized by various diagnostic techniques including those based on the usage of neutron-producing samples. Several independent diagnostics showed that deuterons were accelerated to the unprecedented energy exceeding 30 MeV. This is fifty times the ion energy provided by the generator driving voltage of 0.6 MV and also the highest energy observed in z-pinches and dense plasma foci. High ion energies enable us to obtain unique information about ion acceleration mechanism and to increase the neutron yield. When the 35 mm diameter neutron-producing sample was placed onto the axis below the cathode mesh, the neutron yield was increased up to $(1.1 \pm 0.3) \times 10^{13}$. Considering a $\sim 130 \text{ kJ}$ energy input into the z-pinch, this implies the neutron production efficiency of $\sim 10^8$ neutrons per one Joule of plasma and magnetic energy. The explanation of unique results obtained on GIT-12 may lie in a high impedance of the generator which allowed us to inject plasma shell at a large radius (i.e. to create homogeneous initial conditions before the implosion started) and to disrupt the current-carrying column along a relatively long length.

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